Additional Presentation Materials

Sample of

1. Progress Presentation slides
2. Poster
3. Brochure
4. Video
5. Journal Paper



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Progress Presentation Slides

Week 8- Week 14



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COVER PAGE

MANAGEMENT SYSTEM AND DATA VISUALIZATION FOR SECURITY DEPARTMENT IN DEWAN BAHASA PUSTAKA (DBP)

NAME: IKRAM SYAKIR BIN MOHD AMIN
STUDENT ID: 2016721201

SUPERVISOR: DR. NUR SUHAILAYANI BINTI SUHAIMI



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LCM

Lean Canvas Model

Problem
- Redundant appointment booking due to multiple platform.
- Lack of notification features regarding promotion.
- Difficult to recommend suitable treatment based on patient criteria.



Solution
- Develop a system to manage appointment bookings, provide personalized treatment recommendations, and deliver timely promotion updates for Ilaz Wellness Center.



Key Metrics
- Number of registered patient account.



Unique Value Propositions
- A system that offers personalized treatment recommender by using rule-base.
- Additional notifications feature regarding any updated promotions.



Unfair Advantage
- A pioneering system for personalized treatment recommendations in Homeopathy Electronic Medication.



Channel
- Websites
- Link from Social Media.



Customer Segments
- Homeopathy practitioners in the Electronic Medication.
- Hemopathy in Electronic Medication wellness center.



Cost Structure
- Internet charges.
- System maintenance
- System development
- Database management and maintenance
- Security Services



Revenue Stream
- Membership plan
- Registration fee

DR. NUR SUHAILAYANI SUHAIMI
PENSYARAH KANAN
FAKULTI SAINS KOMPUTER DAN MATEMATIK
UiTM CAWANGAN MELAKA KAMPUS JASIN





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STAKEHOLDER BACKGROUND

Dewan Bahasa dan Pustaka (DBP) is a government building that located at Jalan Dewan Bahasa, Kuala Lumpur.

Implement language and literature planning including the development of language and literature.

DBP Malaysia was first established as Balai Pustaka in Johor Bahru on 22 June 1956.



STAKEHOLDER BACKGROUND

SECURITY DEPARTMENT

- For the project that is going to be developed, it is mainly for security department at DBP.
- Responsible in ensuring that DBP safety level is at its best which includes office safety, property and DBP citizen.
- Perform the manual procedure to manage the keys which records all the key flows in and out from the security room.
- Staff need to fill in the form which are complaint, permission to park a vehicle and permission to borrow access key during weekend that are provided and send it back to security department.
- Based on current situation, web-based and mobile web application software are the one that is proposed to be developed.

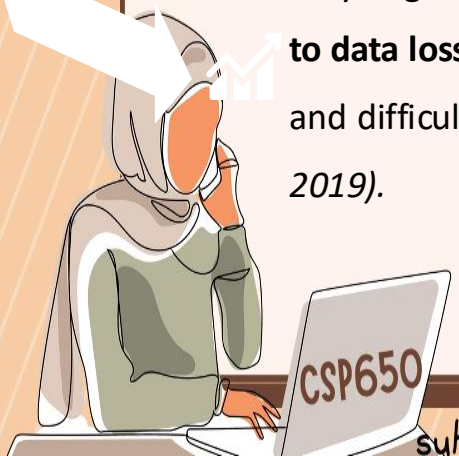


PROBLEM STATEMENTS

Difficulty to manage the manual procedure which is flows in and out keys and still use the paper form that can cause **data loss and time consuming**. *(Noor Lidza, M.T , personal communication, February 21, 2019)*

After staff filled in the permission form to borrow the keys during weekend, the head of department is lack of **time and mobility** for them to approve the application form. *(Hisamudin, M.M, personal communication, February 21, 2019)*

When staff submit the form which is complaint and park a vehicle to the security department, they might have highly chances for them to misplaced and loss the forms which eventually **lead to data loss that can impaired the data visualization** for stakeholder to view particular problem and difficult for them make decision. *(Hisamudin, M.M, personal communication, February 21, 2019).*



AIM & OBJECTIVES

AIM

To develop Management System and Data Visualization for Security Department in Dewan Bahasa Pustaka (DBP).



Objectives

1

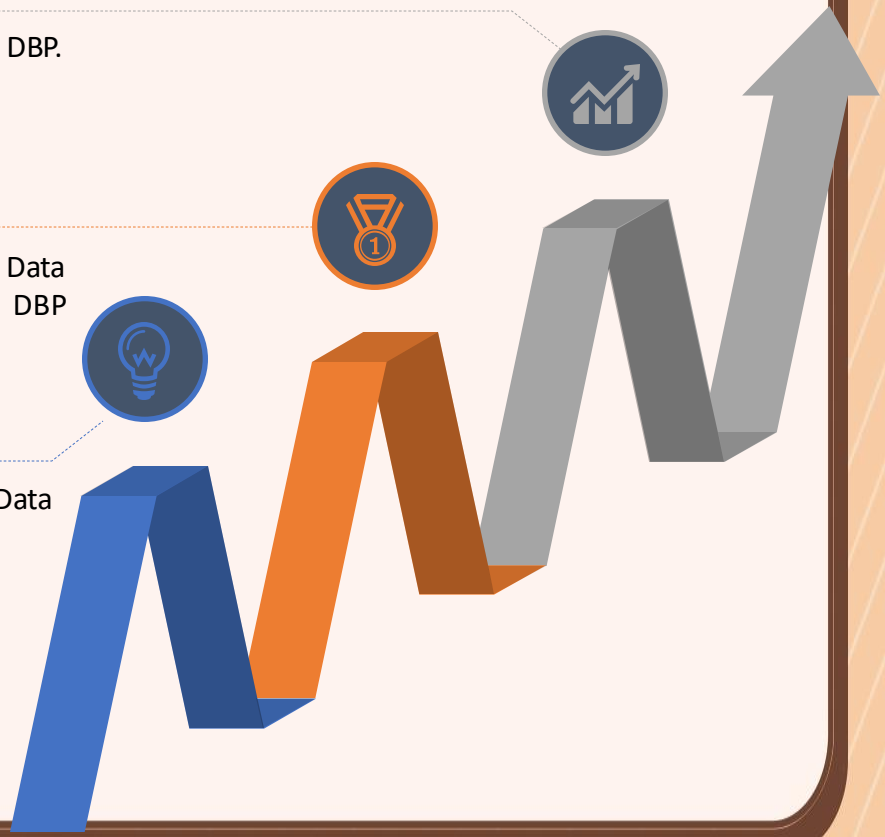
To gather and analyse the requirements from DBP.

2

To design Management System and Data Visualization for Security Department in DBP based on obtained requirements.

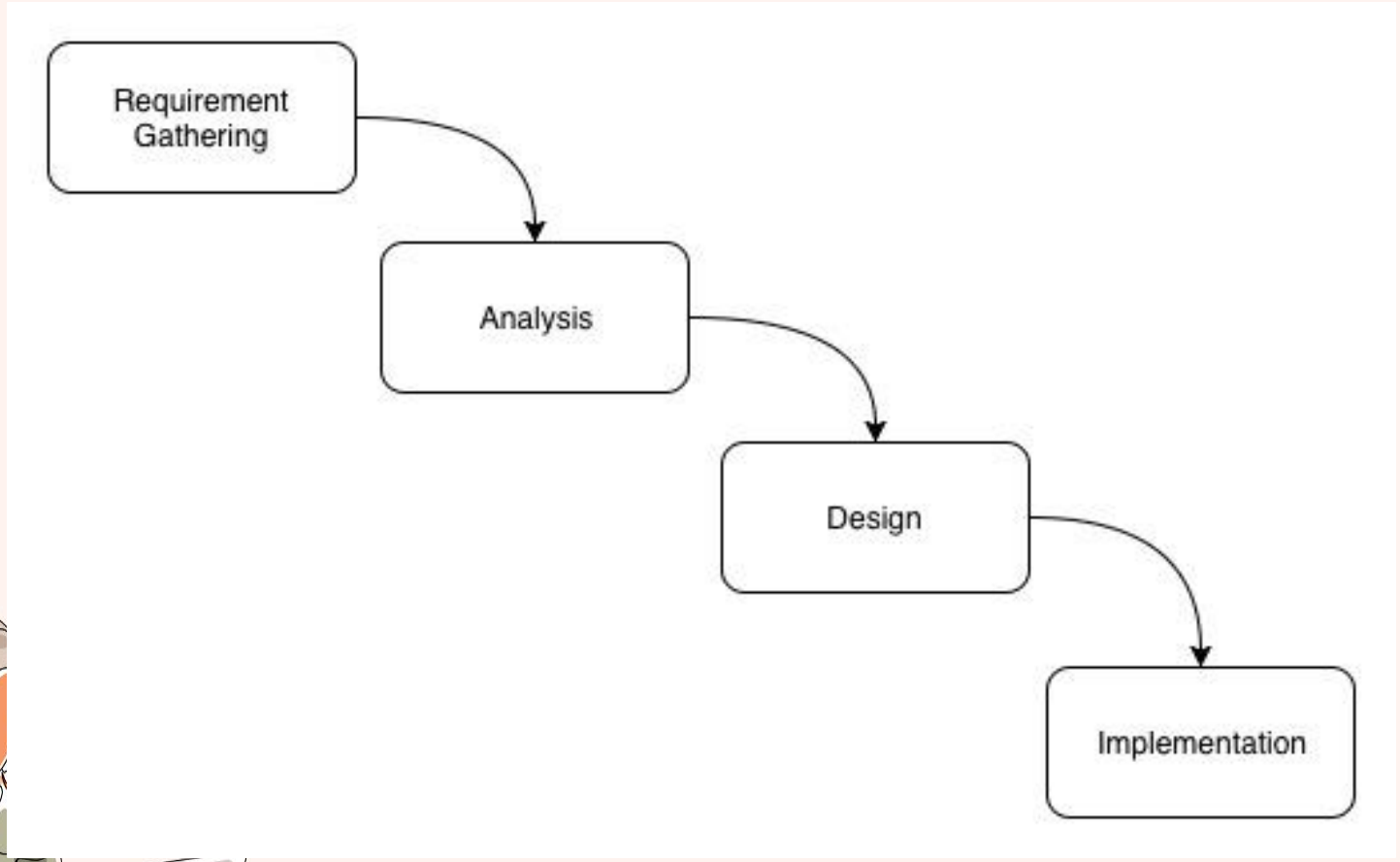
3

To develop Management System and Data Visualization for Security Department in DBP.



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METHODOLOGY



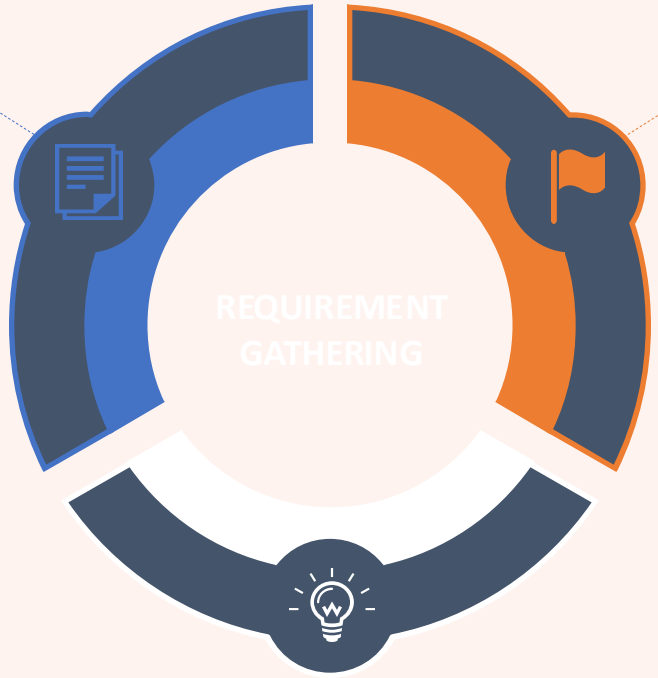
Phases in modified Waterfall Model



METHODOLOGY

ACTIVITIES

1) Prepare a set of questions
2) Interview session with DBP staff
3) Data gathering regarding keys and staff information.



DELIVERABLES

1) A set of interview questions
2) Current business process was identified.
3) Key request procedures together with staff details was identified.

OBJECTIVES

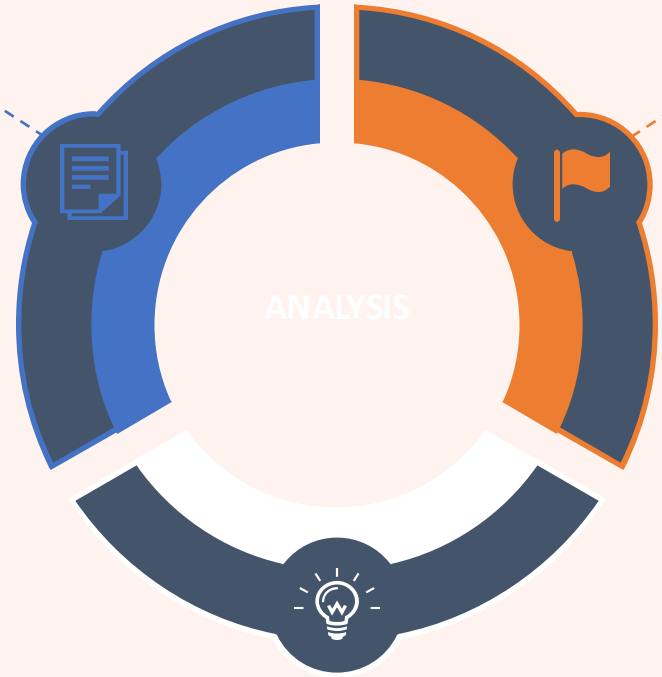
To gather and analyse the requirements from Dewan Bahasa Pustaka.



METHODOLOGY

ACTIVITIES

1) Analyse requirement obtain from interview session with DBP staff
2) Construct diagram activity that shows analysis of requirements
3) Document the requirements



DELIVERABLES

1) Use Case Diagram (UCD)
2) Activity Diagram, Domain Class Diagram (DCD), Sequence Diagram (SSD)
3) Software Requirements Specifications (SRS)

OBJECTIVES

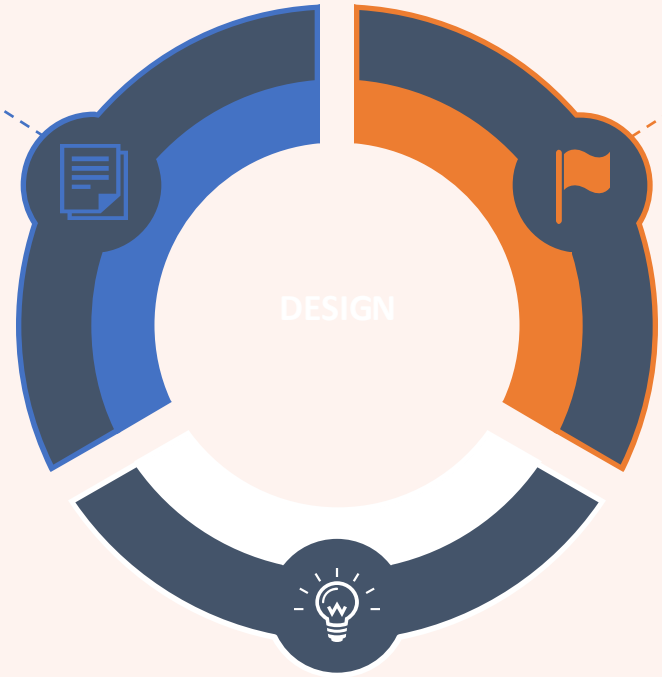
To gather and analyse the requirements from Dewan Bahasa Pustaka.



METHODOLOGY

ACTIVITIES

1) Design storyboard of proposed system
2) Design database
3) Design architecture of system
4) Document the system design



DELIVERABLES

1) Storyboard of the user interface
2) Entity Relationship Diagram (ERD)
3) Multilayer Sequence Diagram (MLSD)
4) Software Design Document (SDD)

OBJECTIVES

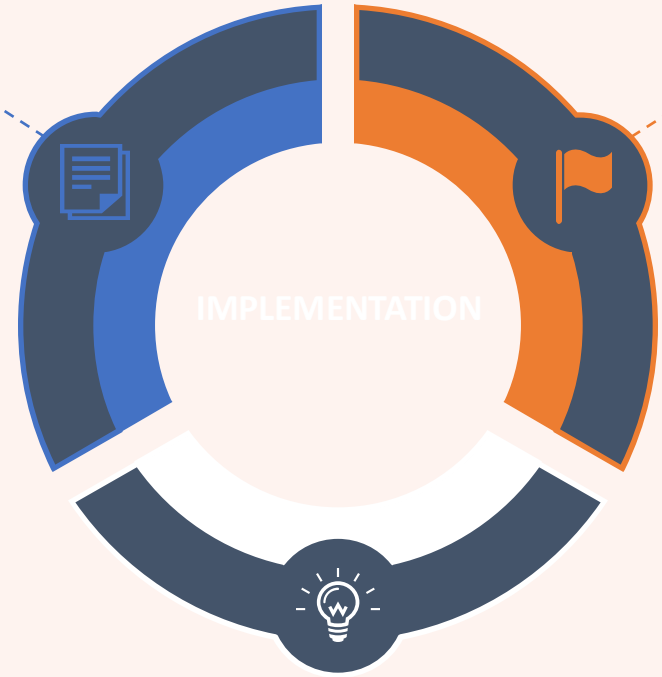
To design Management System and Data Visualization for Security Department in DBP based on obtained requirements



METHODOLOGY

ACTIVITIES

1) Create database
2) Coding the proposed system
3) Create a suitable chart



DELIVERABLES

1) Database architecture
2) Prototype of the system.
3) Using Google Chart for visualization

OBJECTIVES

To develop Management System and Data Visualization for Security Department in DBP



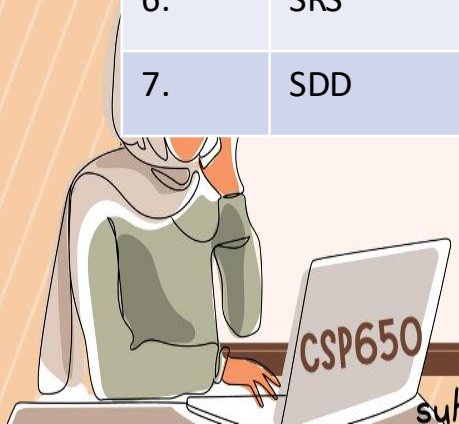
TIMELINE

NO.	USE CASE	STATUS
1.	CREATE STAFF ACCOUNT	COMPLETED
2.	UPDATE STAFF ACCOUNT	COMPLETED
3.	CREATE KEYS INFORMATION	COMPLETED
4.	MANAGE KEY STATUS	COMPLETED
5.	CREATE COMPLAINT	COMPLETED
6.	UPDATE COMPLAINT	COMPLETED
7.	REQUEST PARKING SPACE	COMPLETED
8.	UPDATE PARKING REQUEST	COMPLETED
9.	REQUEST KEY	COMPLETED
10.	UPDATE KEY REQUEST	COMPLETED



TIMELINE

NO.	DOCUMENTATION	STATUS
1.	CHAPTER 1	COMPLETED
2.	CHAPTER 2	COMPLETED
3.	CHAPTER 3	COMPLETED
4.	CHAPTER 4	IN PROGRESS
5.	CHAPTER 5	IN PROGRESS
6.	SRS	IN PROGRESS
7.	SDD	IN PROGRESS



10 WEEK PROGRESS

CSP650 Faculty of Computer & Mathematical Sciences

F5 – PROPOSAL/PROJECT IN-PROGRESS FORM

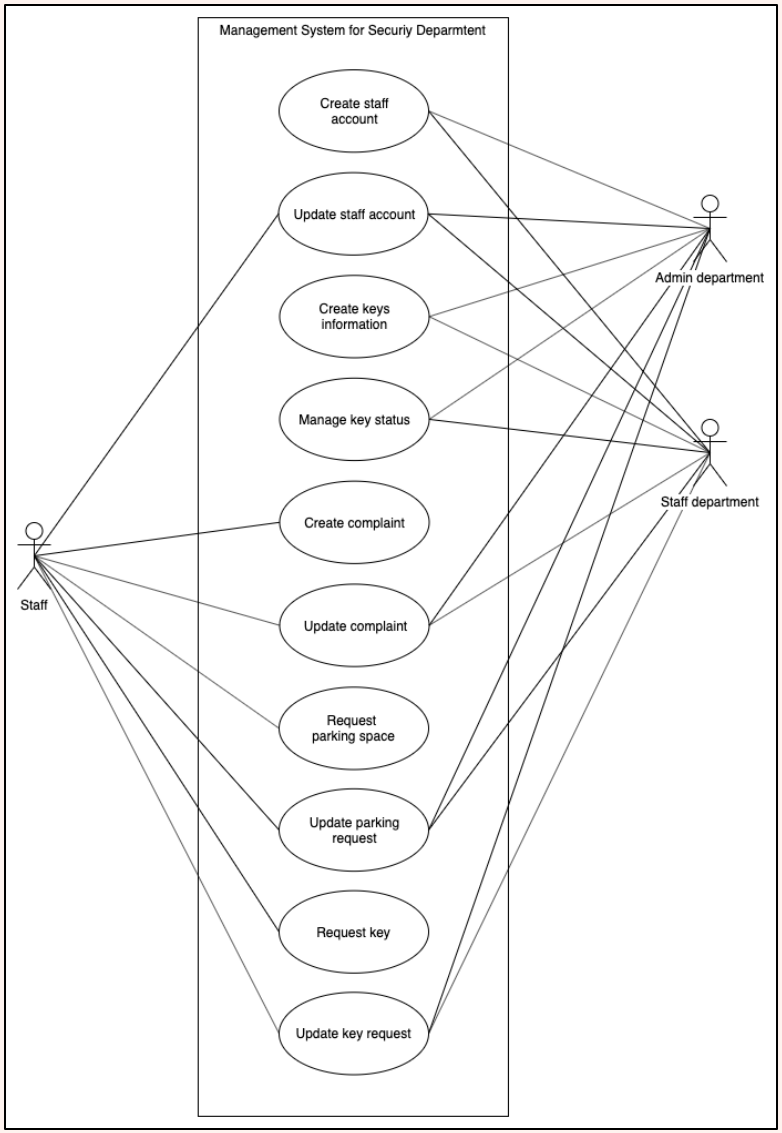
STUDENT NAME	SYAMIMI AFIQAH BINTI SYAHRIL MUKHZANI	**STUDENT ID**	2023380125
PROGRAM	CDCS266		
SUPERVISOR	DR NUR SUHAILAYANI BINTI SUHAIMI		
TITLE	FoodShare: Connecting Surplus to Need, Reducing Food Waste within UiTM Jasin		

DATE OF MEETING	COMPLETED ACTIVITY	SUPERVISOR/CO-SUPERVISOR	
		NEXT ACTIVITY/COMMENT	SIGNATURE
~~18/04/2024~~ 14/10/2024	~~F2 Assignment~~ System interface	UCD (updated) Login page succeed	
22/10/2024	UCD updated & login page succeed	CRUD + updated UCD	
28/10/2024	CRUD cafe & student , CRUD	Notify function	
5/11/2024	Notify function (trial)	Chapter 4 manage leftover	
28/11/24	Chapter 4 & manage leftover	Draft chapter 5	
3/12/24	Draft chapter 4 5 & lean canvas model	Googleform question system	
	~~Chapter 4 & google form question~~		

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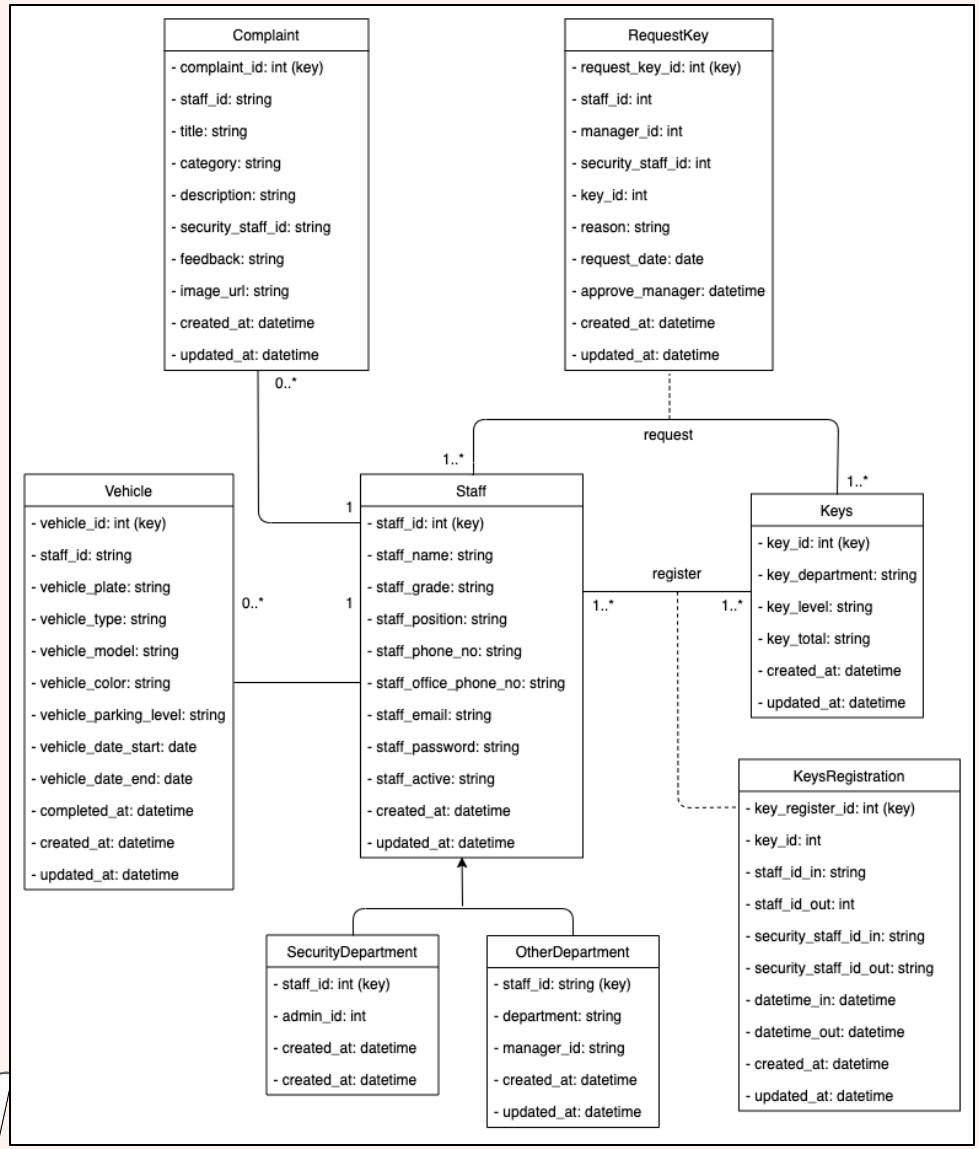


USE CASE


Management System for Securiy Deparmtent
Create staff account
Update staff account
Create keys information
Manage key status
Create complaint
Update complaint
Request parking space
Update parking request
Request key
Update key request
Admin department
Staff department
Staff


CSP650

DOMAIN CLASS DIAGRAM





SYSTEM DEMO



Open for Suggestion and Recommendation
Thank you.





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Item to bring during progress presentation

1. Slides (softcopy in personal laptop)
2. System (personal laptop)
3. Printed slides (2 slides per page)
4. Printed UCD
5. Printed DCD
6. Printed F7



REMINDER!

- Presenter must wear proper attire according to UiTM *Sahsiah Rupa diri*
- Presenter must wear student card
- Presenter must prepare the laptop (with system) 30 mins before first presented started.
- No marks for those without any system.
- ALL students must attend all class session ON TIME even they are not presenting



Material for
FYP Booth Exhibition

1. Your Laptop/Devices
2. Poster
3. Brochure
4. Booth Decoration
5. Video & Journal Paper



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Poster examples



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MANAGEMENT SYSTEM AND DATA VISUALIZATION FOR SECURITY DEPARTMENT IN DEWAN BAHASA PUSTAKA (DBP)

INTRODUCTION

Dewan Bahasa and Pustaka (DBP) is the government organization responsible for regulating the use of the Malay language as well as literature.

This project focuses on Security Department of DBP. This project also will help staff in Security Department in daily operation.

PROBLEM STATEMENT

- Missing paper form that can cause data loss and time-consuming
- Lack of time and mobility to approve the form
- Lack of data that may impair the data visualization

OBJECTIVES

- To gather and analyse the requirements from Dewan Bahasa Pustaka.
- To design Management System and Data Visualization for Security Department in DBP.
- To develop Management System and Data Visualization for Security Department in DBP.

METHODOLOGY



RESULT



SIGNIFICANCE

1. Information will be stored in more consistent database.
2. Successfully reduce time-consuming.
3. Able to make decision based on generated report.

CONCLUSION

The management system is suitable to handle the workload in security department and also for the staff in DBP. By using data visualization, stakeholder can view a particular problem to make the decision.



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Final Year Project

Prediction System to Manage Inventory for AAIRA using Naive Bayes Technique

UNIVERSITI TEKNOLOGI MARA

INTRODUCTION

Prediction System to Manage Inventory for AAIRA using Naive Bayes Technique is develop to minimize the problems encountered by online seller, AAIRA. The aim of this project is to develop a web-based prediction system to assist the process of predicting the class of the products when restock based on customers' preferences. The outcome of this prediction system is to display the list of the predicted products with the calculated probability. The suggested products for promotion is based on the result of prediction.

PROBLEM STATEMENTS

- Difficult to trace the customer preferences.
- Lack of knowledge in data analysis for marketing of the product.
- Abundance of products in store.

OBJECTIVES

- To gather data and analyse requirements from online seller, AAIRA.
- To analyse data using suitable data mining technique.
- To develop the prototype of the prediction system based on the analysed data.

METHODOLOGY



RESULTS



Prediction Page



Promotion Page

CONCLUSION

The development of the prediction system is able to assist the online seller, AAIRA in managing the inventory efficiently by using the analysis of naive bayes technique and knowledge discovery of database methodology.





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ONLINE ZAKAT MANAGEMENT SYSTEM FOR UiTM JASIN USING RULE BASED TECHNIQUE

INTRODUCTION

- Zakat ACIS in UiTM Malacca Jasin Campus
- Zakat applicants consists of FSKM and FPA students
- Current processes have problems in managing the zakat such as:

. Missing data of zakat application form
. Time consuming for staff to manage the form
. Difficulties to filter students qualification based on zakat requirements.



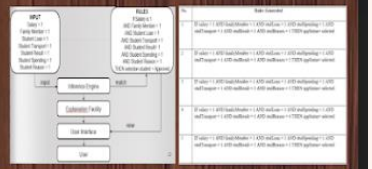
AIM:

To develop an Online Zakat Management System for UiTM Jasin Using Rule Based Technique

OBJECTIVES:

. To gather and analyze the requirement based on collected data.
. To design the system using Rule Based Technique.
. To develop the system with embedded constraint rules.

METHODOLOGY:



LIMITATIONS:

1) For UiTM Jasin students only
2) Does not cater on zakat transaction
3) Limited inference engine

RESULT AND FINDINGS:







CONCLUSION:

Online Zakat Management System for UiTM Jasin using Rule Based Technique is a web-based system that assist staff to filter student's qualification with zakat requirements efficiently. This project will allow ACIS staff to approve students qualification based on the input entered by students. Hence, by implementing rule based technique, it will assist the staff in reducing time to manage the student's application and enhance staff productivity as management of zakat is getting easier. For future enhancement, it is suggested to add more inference engine and provide a notification through email to student.



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Flood Forecasting In Malacca River Basin Using Artificial Neural Network (ANN)

Introduction

The study area of this project focuses on river basin where it is a land that water flows across or under on its way to a river. Malacca consists of four major river basin ; Kesang river basin, Linggi river basin, Duyong river basin and Malacca river basin. In addition, December 2006 was the year where Malacca experienced the worst flood ever recorded by DID Malacca. In order to forecast future flood, Artificial Neural Network (ANN) is applied into this research.

Aim

To demonstrate the accuracy of Artificial Neural Network (ANN) as a flood forecasting tool for Malacca river basin in order to prevent future disaster.

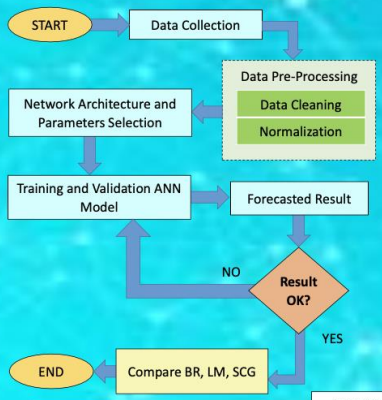
Problem Statement

1. Huge amount of data – difficult to analyze manually
2. *Infobanjir* and *Publicinfobanjir* only focused on providing alert or as flood warning system
3. Lack of study areas attempted in Malacca river basin for flood forecasting

Objectives

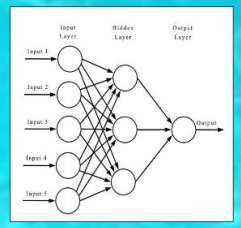
- To collect and analyze the data gathered from Water Resource Management And Hydrology Division, Department of Irrigation And Drainage (DID) Malaysia and Angkatan Pertahanan Awam (APM) Malacca
- To design Artificial Neural Network (ANN) framework and methods
- To compare Artificial Neural Network (ANN) which is suitable for flood forecasting in Malacca river basin.

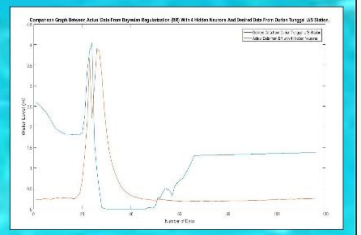
Methodology

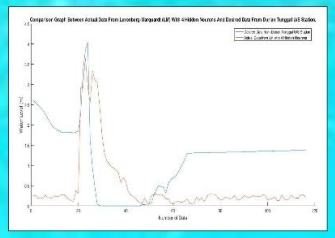


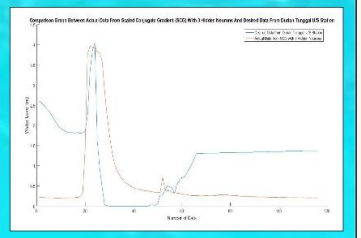
Result & Findings

ANN Training Algorithm (No. Hidden Neurons)	RMSE	R^2	Comparison Of Average Value For Water Level at Highest Peak In Testing Range (Between 20th Data Until 29th Data)				
			Actual Data From Forecasting Activities	Desired Data From Durian Tunggal U/S Station	Net	Error Accuracy	Total Accuracy
BR (4)	0.02698	0.99	2.643595	1.841	0.80295	0.44	56 %
LM (4)	0.019844	0.99	2.726274	1.841	0.885274	0.48	52%
SCG (3)	0.03451	0.94	3.338391	1.841	1.497391	0.81	19%









Conclusion

In summary, Bayesian Regularization (BR) with 4 hidden neurons produce the best performance among the three ANN training algorithms throughout ANN training as well during its testing and thus, make it as the most suitable to use as ANN training algorithm for flood forecasting in Malacca river basin.





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TRADITIONAL AND COMPLIMENTARY TREATMENTS RECOMMENDATION SYSTEM USING RULE-BASED CLASSIFIER

INTRODUCTION

AppleRose Healing Centre (ARHC) is a treatment centre that provides traditional and complimentary medication and treatments for all possible medical cases. The main service offered are cupping (bekam), acupuncture and reflexology treatment depending on the patient conditions. Due to unending demand from customers, several problems have come upon the organization that requires for automated service.

PROBLEM STATEMENTS

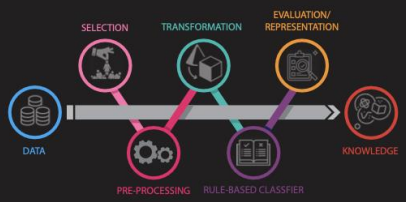
- Manual analysis requires a lot of time due to having to keep referring to guidelines.
- Human is prone to mistakes. Decision is made based on human judgement depending on condition of practitioners.
- Time consumption issue disrupts the number of patients that can be handled per day.

OBJECTIVES

- To gather and analyze data from ARHC
- To design the recommendation system based on the analyzed data findings.
- To develop a recommendation system for Apple-Rose Healing Centre.

METHODOLOGY

Knowledge Discovery in Databases (KDD)



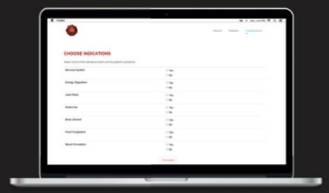
AIM

To develop a recommendation system using rule-based classifier which eases the process of selecting the correct treatments based on health indications from the patients.

RESULT & FINDINGS

Use Case Diagram (UCD)









CONCLUSION

Treatment Recommendation System aids in making a better decision on what type of treatments shall the practitioners deem suitable for associated patients. With the use of this system, it reduces the constraints of time consumption, increases reliability towards the business practice as well as the productivity level of the Apple Rose Centre.



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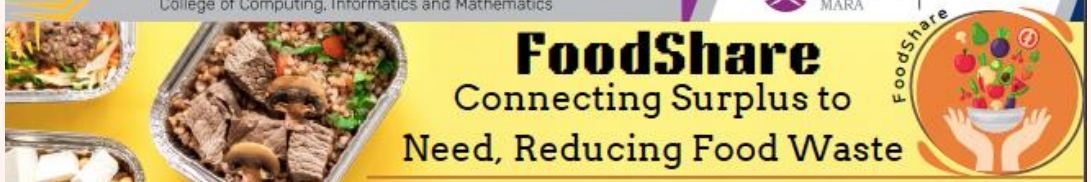


Kolej Pengajian Pengkomputeran, Informatik dan Matematik
College of Computing, Informatics and Mathematics

Universiti Teknologi MARA
Cawangan Melaka
Kampus Jasin

FoodShare

Connecting Surplus to Need, Reducing Food Waste



INTRODUCTION

FoodShare is a system to manage leftover food distribution and is beneficial in reducing the problem of food waste and food insecurity. This system is developed for the use of cafeteria owners and students at UiTM Jasin.

KEY PROBLEMS

- Difficulties in sharing the leftover information with students
- Ineffective distribution of leftover meals among the students
- Lack of booking features for event

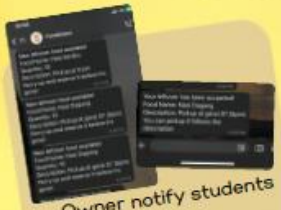
METHODOLOGY

Requirement and Analysis → Design → Implementation

OBJECTIVES

1. To gather data and analyze requirements of FoodShare
2. To design the system of FoodShare
3. To develop a system of Foodshare

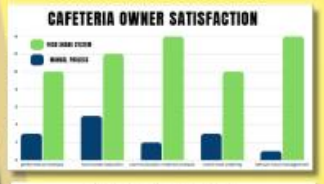
FINDINGS

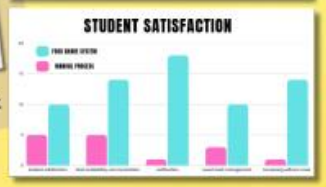


Owner notify students



Booking Management





CONCLUSION

82.05% efficiency yielded from the FoodShare usage of cafe owner and 82.42% students satisfaction produced.



Brochure examples



RESULT

	Inputs Entered	Rules (constraint) matched? (%)	Result
Student A	salary = 1 dependent = 1 studLoan = 1 studTransport = 1 studPhysical = 1 studSpending = 1 studResult = 1	100%	Approved

SENARAI NAMA PENERIMA ZAKAT 2019

Tahniah kepada pelajar yang layak menerima zakat. Kepada pelajar yang tidak mempunyai nama dalam senarai, nama anda masih berada di senarai menunggu. Anda boleh klik pada nama anda untuk mengetahui kelayakan anda

Ic	900901115744
Nama Pelajar	TUAN NUR ALLIA BINTI TUAN MOHD ZAMLI
Gender	perempuan
Fakulti	fskm
Sem	enam
Negeri	trg
Status	Approved

	Inputs Entered	Rules (constraint) matched? (%)	Result
Student B	salary = 4 dependent = 1 studLoan = 3 studTransport = 1 studPhysical = 1 studSpending = 1 studResult = 1	0%	Waiting list

Maaf. Permohonan anda dalam proses.

Sila KLIK DI SINI bagi kembali ke halaman utama

Klik Di Sini

CONCLUSION

Online Zakat Management System for UiTM Jasin using Rule Based Technique is a web-based system that assist staff to filter student's qualification with zakat requirements efficiently. This project will allow ACIS staff to approve students qualification based on the input entered by students. Hence, by implementing rule based technique, it will assist the staff in reducing time to manage the student's application and enhance staff productivity as management of zakat is getting easier. For future enhancement, it is suggested to add more inference engine and provide a notification through email to student.

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ONLINE ZAKAT MANAGEMENT SYSTEM FOR UITM JASIN BY USING RULE BASED TECHNIQUE





INTRODUCTION

ACIS is one of institution under Universiti Teknologi Mara (UiTM) Malacca Jasin Campus which provides zakat to UiTM students. There are consists of FPA and FSKM students that can apply the zakat. Currently, ACIS staff faced several problems in managing the process of zakat due to the process was being done manually. Therefore, this project is conducted to help ACIS staff to manage student's criteria with zakat requirements provided.

PROBLEM STATEMENTS

- Missing data of zakat application form
- Time consuming for staff to manage the form
- Difficult to filter student's qualification with zakat requirement

AIM:

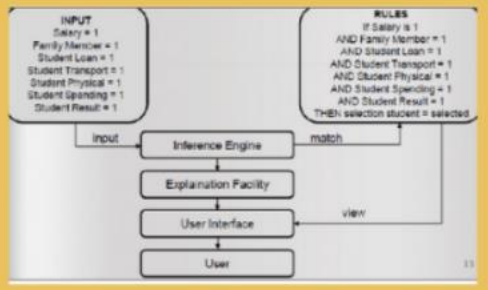
To develop an Online Zakat Management System for UiTM Jasin by Using Rule Based Technique.

OBJECTIVES:

1) To gather and analyze the requirement based on collected data.
2) To design the system using Rule Based Technique.
3) To develop the system with embedded constraint rules.

TECHNIQUE

Rule Based Architecture



No.	Rules Generated
1	IF salary = 1 AND familyMember = 1 AND studLoan = 1 AND studSpending = 1 AND studTransport = 1 AND studResult = 1 AND studReason = 1 THEN appStatus= selected
2	IF salary = 1 AND familyMember = 1 AND studLoan = 1 AND studSpending = 1 AND studTransport = 1 AND studResult = 1 AND studReason = 2 THEN appStatus= selected
3	IF salary = 1 AND familyMember = 1 AND studLoan = 1 AND studSpending = 1 AND studTransport = 1 AND studResult = 1 AND studReason = 3 THEN appStatus= selected
4	IF salary = 1 AND familyMember = 1 AND studLoan = 1 AND studSpending = 1 AND studTransport = 1 AND studResult = 1 AND studReason = 4 THEN appStatus= selected
5	IF salary = 1 AND familyMember = 1 AND studLoan = 1 AND studSpending = 1 AND studTransport = 1 AND studResult = 2 AND studReason = 1 THEN appStatus= selected



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ABOUT UFPSS

This system is developed to help Borneo Headhunters in selecting the players for their hat. Up till now, they are still using manual selection processes to select the players and it consumes a lot of time to do so.

There are a few steps in selecting the players manually. First, the players must register themselves on the Google form prepared by the organizer. After a week, registrations will be closed and the organizers will start to look into every registrations. Then, they will group the players according to their ultimate skills. From there, they start to eliminate and select players.

The method and technique that is used to develop this system is SDLC waterfall and rule based technique.







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THE ULTIMATE FRISBEE PLAYER SELECTION SYSTEM (UFPSS) FOR BORNEO HEADHUNTERS USING RULE BASED TECHNIQUE



PROBLEM STATEMENT

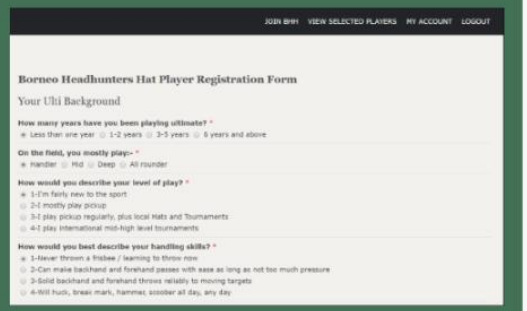
1. Human Judgment
2. Time Consuming
3. Tedious Process

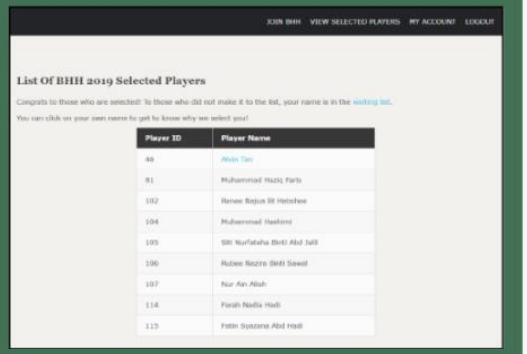


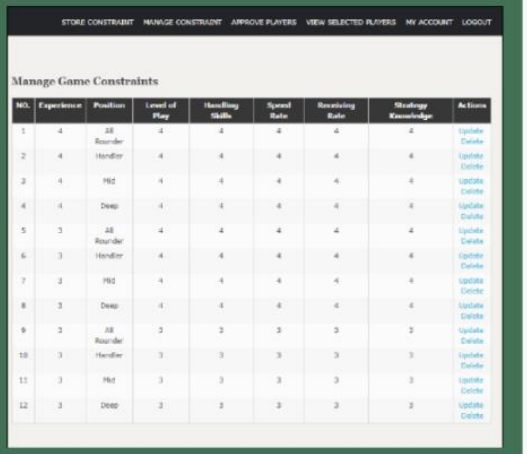
AIM

To produce a web-based system that can assist the tournament organizer in selecting the players through a systematic and fair processes.

RESULTS







OBJECTIVES

1. To gather and analyze data from Borneo Headhunters.
2. To generate the rules for UFPSS.
3. To design and develop the prototype by using the generated rules.

CAPABILITIES OF UFPSS

ADMIN

1. Store Game Constraint
2. Manage Game Constraint
3. Approve Applications
4. View Selected Players

PLAYER

1. Register Account
2. Manage Account
3. Apply Tournament
4. View Selected Players



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CONCLUSION

The development of the prediction system is able to assist the online seller, AAIRA in managing the inventory efficiently by using the analysis of Naive Bayes technique and knowledge discovery of database methodology.



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UNIVERSITI TEKNOLOGI MARA

PREDICTION SYSTEM TO MANAGE INVENTORY FOR AAIRA USING NAÏVE BAYES TECHNIQUE





PROBLEM STATEMENTS

- Difficult to trace the customer preferences.
- Lack of knowledge in data analysis for marketing of the product.
- Abundance of products in store.

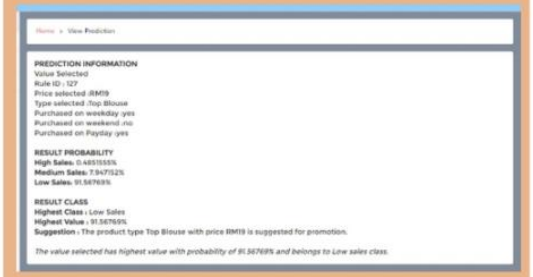
OBJECTIVES

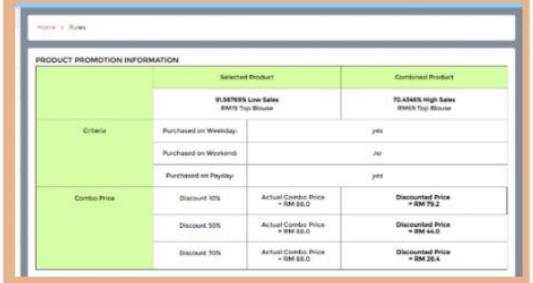
- To gather data and analyse requirements from online seller, AAIRA.
- To analyse data using suitable data mining technique.
- To develop the prototype of the prediction system based on the analysed data.

METHODOLOGY



RESULT





SIGNIFICANCES

STAKEHOLDER

a. The online seller will manage the process of ordering products in more efficient manner.

b. Less data redundancy or loss of important information as all of the important information is stored in more reliable database.

c. This system will assist the online seller in analysing and predicting their inventory in conjunction with the current market demand.

d. Less time consuming in predicting the products based on previous order as the prediction involved data mining technique.

ACADEMICIAN

a. Help and assist students, lecturers and staffs in terms of reference purposes where analysis of the related studies can be obtained from this thesis and act as a guideline.

b. Academicians will be able to explore and gain more knowledge in data mining technique that will aid the process of analysis of this thesis.

c. Academicians are able to develop self-learning process as the process of using data mining technique involves academicians to collect and gather data, analysis the suitable data, turn the data into knowledge and information through detailed research which



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Booth Decoration

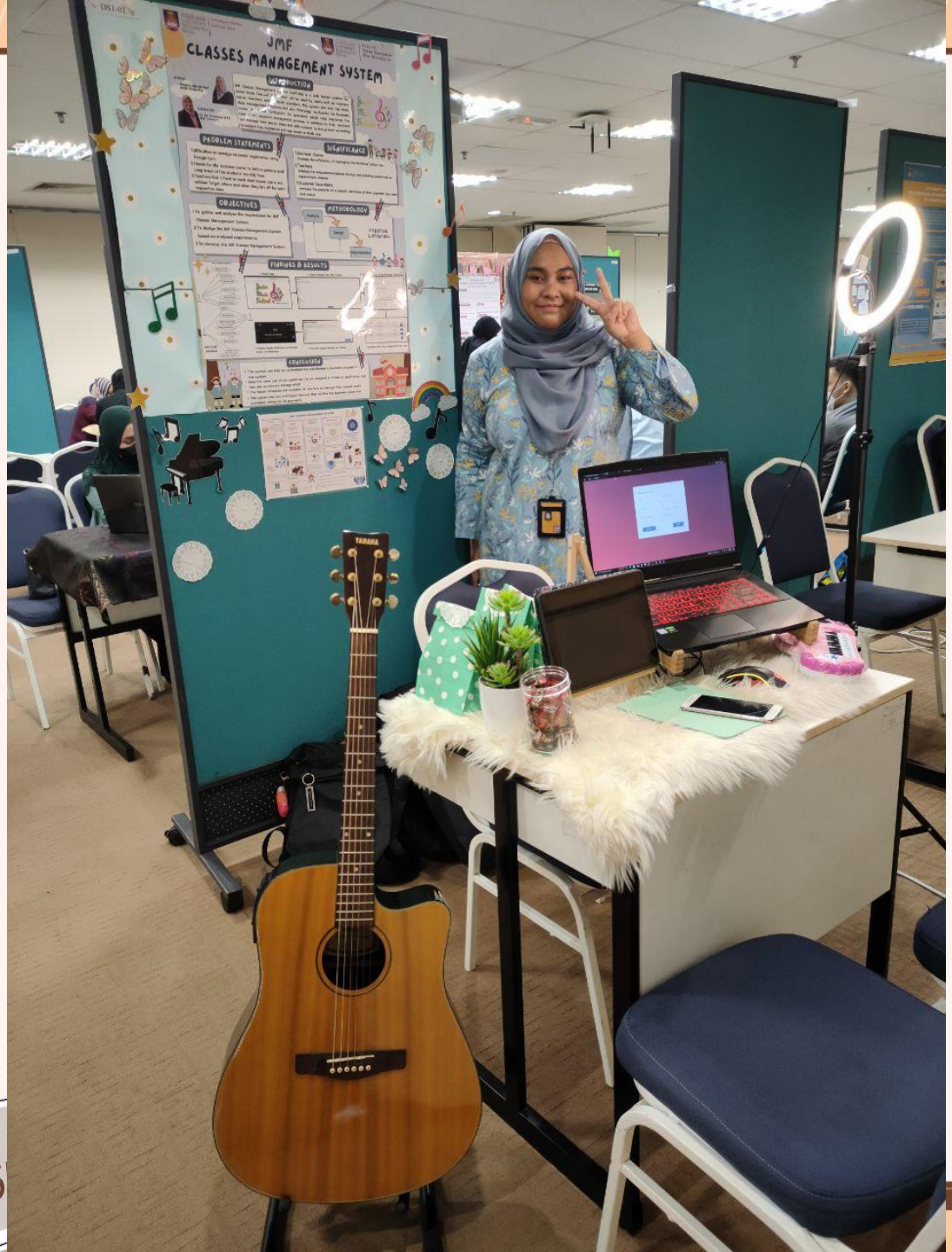


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METHODOLOGY


CSP6


PALAM TECH PC BUILD GUIDER
PALAM TECH PC BUILD GUIDER
with recommendation system
INTRODUCTION
PROBLEM STATEMENT
OBJECTIVES
METHODOLOGY
RESULT
CONCLUSION
Build your dream PC
Welcome, Arif azhar


JMF
CLASSES MANAGEMENT SYSTEM
INTRODUCTION
PROBLEM STATEMENTS
SIGNIFICANCE
OBJECTIVES
METHODOLOGY
FINDINGS & RESULTS
CONCLUSION
YAMAHA


CSP6


RTE
DS8 - 22
UNDER CONSTRUCTION
KPPIM
JULY 2024
DOSNANI IBRAHIM ENTERPRISE

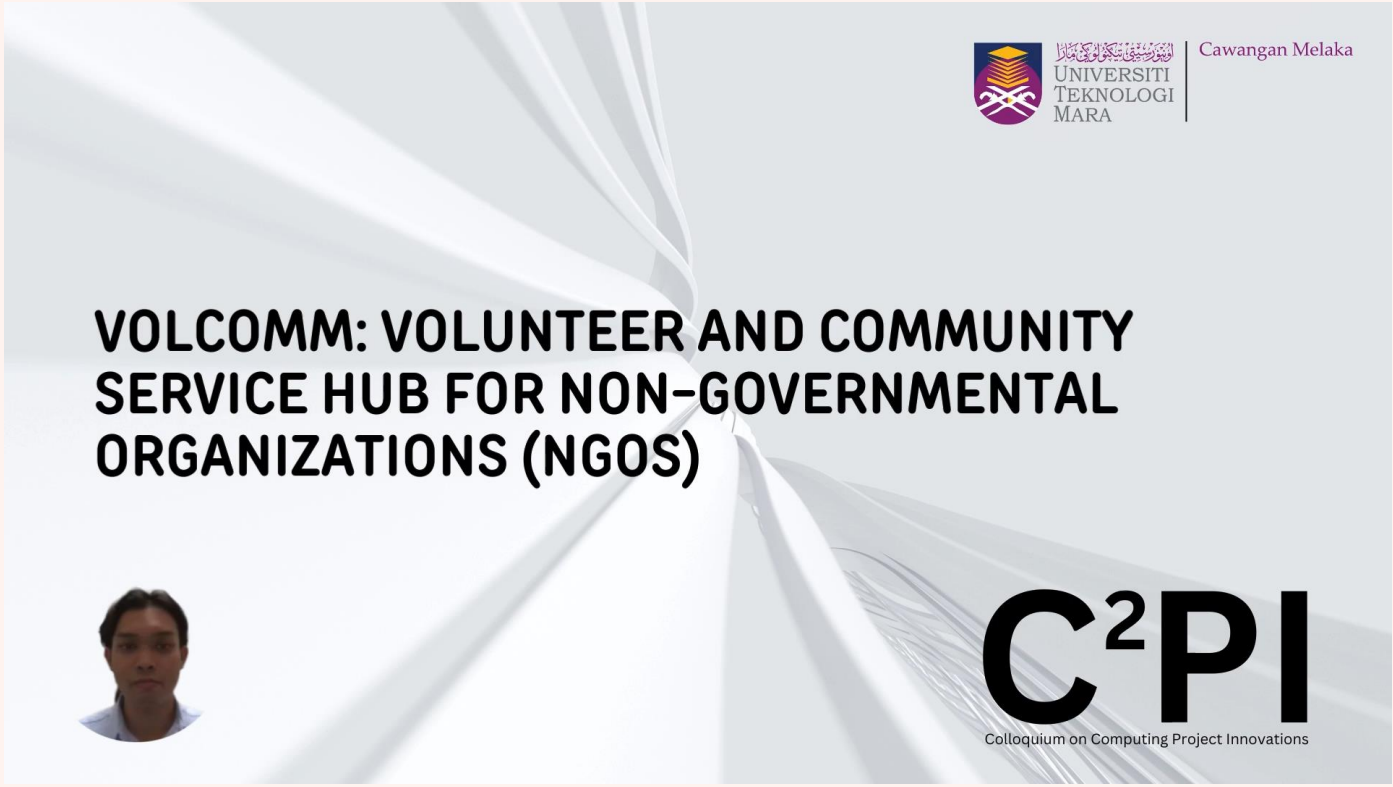

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FINDINGS
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KPPIM
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cooking class management and evaluation system


CSP6

VIDEO SAMPLES



VIDEO SAMPLE #01





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Progress in Computer and Mathematics Journal (PCMJ)
volume 1, issues 1 [March, 2024] pp. x
e-ISSN: XXXX-XXXX
Website:

VOLCOMM: VOLUNTEER AND COMMUNITY SERVICE HUB FOR NON-GOVERNMENTAL ORGANIZATIONS (NGOs)

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Article Info

Received: 1st February 2024
Accepted:
Available Online:

Abstract

VolComm is a web-based platform designed to assist NGOs and volunteers in efficiently managing community service activities. Volunteers can register, create profiles, and discover events aligned with their interests and skills, while NGOs can post opportunities, track participation, and generate digital certifications. VolComm incorporates smart matching to align volunteers with appropriate activities and provides automated notifications. A survey revealed that 87% of users found VolComm effective in managing events, while 92% reported that it streamlined the process of finding volunteer opportunities. Future enhancements include adding more volunteer categories, integrating geolocation for nearby event discovery, and developing a donation module for NGOs. These improvements aim to make VolComm a more user-friendly and impactful platform for volunteer engagement.

Keywords: Volunteer Management System, NGO, Digital Certification, Web-Based Platform, Community Service

INTRODUCTION (HEADING 1)

Volunteer organizations play a crucial role in addressing societal needs. However, existing platforms often lack a centralized system to match volunteers with opportunities effectively. Group Volunteer Malaysia (GVM), a Facebook group, currently serves as a medium for NGOs to post opportunities, but it lacks filtering features, efficient volunteer management, and certificate issuance. Volunteers struggle to find relevant opportunities due to the lack of structured event categorization and automated matching features.

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Progress in Computer and Mathematics Journal (PCMJ)
volume 1, issues 1 [March, 2024] pp. x
e-ISSN: XXXX-XXXX
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Additionally, the absence of a certification system makes it difficult for volunteers to maintain records of their contributions. This results in challenges for organizations that need to track engagement and recognize volunteers effectively.

To overcome these issues, VolComm is designed as a web-based system to streamline volunteer management, event organization, and certification tracking. The platform aims to improve the volunteer experience by providing personalized recommendations, automated notifications, and centralized certificate management.



Figure 1: Flowchart Volunteer Registration on Facebook

Figure 1 illustrates how organizations and volunteers can link up via Facebook. Organizations post volunteer opportunities on the group, and volunteers search through these listings to identify suitable matches. Once the volunteer found an opportunity that catches their attention, they fill the application form that are stated on the post. The organization evaluates the application and determines if they will approve or reject it. If the volunteer is approved, the organization will get in touch to move forward. If the volunteer is not accepted, the process will stop for them. Volunteers who are accepted may also be contacted by the organizer.

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LITERATURE REVIEW

Managing volunteer activities efficiently remains a challenge for many organizations, often leading to difficulties in recruitment, communication, and record-keeping. Traditional methods, such as social media groups or manual registrations, can be disorganized and time-consuming. VolComm was developed to streamline volunteer management by simplifying the process of finding opportunities and registration. To support the development of VolComm, this literature review examines existing volunteer management systems, the role of technology in improving event coordination

Volunteering and Community Services

According to Eyler and Giles (1999), community service happens when people help others or their community without getting paid. Different types of services, like charity care, health promotion, research and education, and service accessibility, are offered by these organizations (Ginn, Shen, & Moseley, 2009). Singh, Young, Loomer, and Madison (2018) stated that Nonprofit hospitals have a notable impact on communities by providing benefits like charity care, and performing community health needs assessments.

Volunteering refers to various activities undertaken by individuals during their leisure time to support others, whether through informal assistance or formal organizational involvement (Tucker & Gearhart, 2022). According to Lum and Lightfoot (2005), volunteering has been associated with positive effects on physical and mental health, particularly among older individuals. Mustafa, Radzi, and Hamzah (2020) outlined four primary motivations behind individuals' engagement in volunteering, which are the desire to contribute to the community, the opportunity to acquire new skills, heightened awareness of societal issues, and the expansion of social networks.

Volunteer Management System (VMS)

Volunteer management has evolved to focus on providing tailored practices that suit each organization's unique context, much like how human resource management operates (Piatak & Carman, 2023). Volunteer Management Systems (VMS) play a crucial role in this evolution by simplifying the process of connecting volunteers with opportunities, organizing tasks, facilitating communication, and implementing evaluation and motivation strategies

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(Schonbock, Raab, Altmann, Kapsammer, Kusel, Proll, Retschitzegger, & Schwinger, 2016). According to Chui and Chan (2019), VMS also help organizations expand and diversify their volunteer pool by enabling efficient recruitment, screening, and onboarding of volunteers. VMS are essential across various sectors, including emergency and disaster management, global health training, and nonprofit organizations. Informal volunteers are a valuable resource during emergencies and disasters, underscoring the importance of understanding their roles and challenges (Whittaker, McLennan, & Handmer, 2015). According to MacNairn (2019), in global health programs VMS can foster mutually beneficial partnerships and optimize benefits while minimizing costs or harms. Organizational factors significantly impact volunteer coordination, necessitating a comprehensive approach to enhance volunteer engagement and effectiveness (Studer & Schnurbein, 2012).

Challenges in Volunteer Management

Jacob (2016) highlighted that some volunteers may prioritize their personal satisfaction above organizational objectives. Consequently, they may engage in disruptive behavior, such as breaking rules, particularly when rules or their enforcement impede their enjoyment or personal aspirations. According to Bilan (2019), volunteers frequently encounter challenges stemming from ambiguity surrounding their roles and the corresponding rules, which can result in inconsistent behavior and difficulties in meeting organizational expectations. Moreover, the uncertainty surrounding roles can contribute to high turnover rates, as volunteers disengage due to unclear expectations or dissatisfaction with their volunteer experience. This turnover not only disrupts organizational continuity but also poses challenges in maintaining a committed volunteer community. According to Cho, Wong, & Chiu (2020), these issues of role ambiguity and turnover continue to be significant challenges in volunteer management.

Comparison of Related Works

Volunteer Malaysia Group on Facebook, VolunteerHub, and BetterImpact are all platforms and websites established with the purpose of increasing volunteerism, but the way they work and the capabilities are different. Volunteer Malaysia Group on Facebook, Join this FB group is to help connect all the volunteers and organisations, providing a wider coverage and rather a more casual communication. On the other hand, BetterImpact provides an elaborate volunteer management software with application such as; tracking volunteers,

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analytics, and communication among others. In each of the platforms described above there are pros and cons and therefore deciding which one to use depends on the objectives and approximately on the nature of the activity of an organization or a project. Table 1 describes these platforms and their attributes wherein users can compare and contrast to make their decisions.

Table 1: Comparison of Related Works

Feature	Volunteer Malaysia Group on Facebook	VolunteerHub	BetterImpact
Volunteer Opportunity Listings	Yes	Yes	No
Online Volunteer Registration	No	Yes	Yes
Volunteer Scheduling	No	Yes	Yes
Communication Tools	Limited (Comments, Messenger)	Yes (Emails, Notifications)	Yes (Emails, Messages)
Reporting and Analytics	No	Yes	Yes (Advanced Reporting)
Integration Capabilities	No	Yes (CRM, Email Marketing, etc)	Yes (CRM, Integration)
Customization Options	No	Yes	Yes (Extensive Customization)
Mobile Accessibility	Yes (Facebook App)	Yes (Mobile App)	Yes (Mobile App)

Comparison of Development Methodology

Agile, waterfall, and incremental models are all used in software development, but each has its strengths and weaknesses. Agile is flexible and focuses on customer collaboration, while waterfall follows a linear approach with distinct phases. Incremental combines elements of both, breaking the project into smaller parts. Choosing the best method depends on the project's needs and goals. Table 2 shows the comparison of method which are agile, waterfall, and incremental.

Table 2: Comparison of Development Methodology

Model	Agile	Waterfall	Incremental
Approach	Flexible and iterative	Sequential and linear	Combines elements of Agile and Waterfall

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Flexibility	High	Low	Moderate
Phases	Multiple short cycles	Distinct phases (Requirements, Design, Implementation, Testing, Deployment)	Phased approach with increments
Iterative	Yes	No	Yes

Comparison of Notification services

Email notification, WhatsApp notification, and SMS notification services are all utilized for communication purposes, each presenting its own advantages and drawbacks. Email notifications offer versatility and can convey detailed information, while WhatsApp notifications prioritize real-time interaction and multimedia sharing. On the other hand, SMS notifications provide simplicity and reliability, often reaching users even in areas with limited internet connectivity. The most suitable notification method depends on requirements and objectives of the project. Table 3 presents a comparison of email, WhatsApp, and SMS notification services.

Table 3: Comparison of Notification services

Aspect	Email notification	WhatsApp notification	SMS notification
Delivery Speed	Varies, can be delayed	Real-time	Nearly instant
Accessibility	Requires internet connection	Requires internet connection	Works with or without internet
Message Length	Typically longer	Limited characters	Limited characters
Multimedia Support	Yes, supports attachments	Yes, supports multimedia files	No
User Interaction	Limited interactivity	High interactivity	Limited interactivity
Cost	Often free or low-cost	Free	Often free or low-cost
Reliability	Subject to spam filters	Dependent on internet connection	Generally reliable

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METHODOLOGY

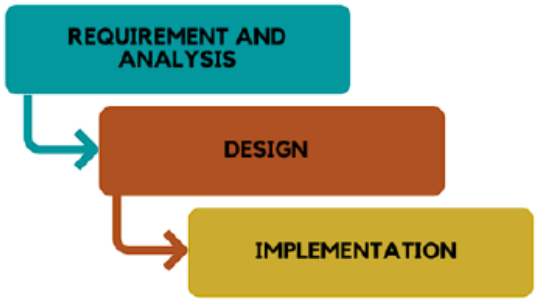


Figure 2: Waterfall Methodology

The development of VolComm followed a clear Waterfall methodology, ensuring that each stage was completed one after the other. This helped make sure that every part of the project was fully planned, checked, and confirmed before moving to the next step. The project started with the Requirement Gathering and Analysis phase. During this stage, surveys and interviews were conducted with volunteers and event organizers to understand the main features needed for the system. This phase showed that important features included an easy registration process, an efficient way for volunteers to find opportunities, and the ability to send real-time notifications. Organizers also highlighted the need for a simple way to manage volunteers and communicate with them. The information collected during this phase laid the groundwork for the rest of the project.

In the System Design phase, technical diagrams were created to show how the system would work. Use Case Diagrams were made to describe how different users would interact with the system. Entity-Relationship (ER) Diagrams were created to organize the data and how it would be stored in the system's database. System Sequence Diagrams were used to show the steps users would take when using the system. These diagrams helped refine the design and made sure it met all the requirements.

In the final Implementation phase, the design was turned into a working system. VolComm was developed using Laravel for the backend, MySQL for the database, and AWS

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for storage. Features like easy registration, advanced search options, event management, and automated notifications were successfully added. The system was thoroughly tested during this phase, including testing by real users to ensure it was easy to use and met expectations. By following the Waterfall methodology, each phase was completed carefully, ensuring the project met all its goals and the final system was fully functional.

RESULT AND DISCUSSION

The survey results provide a comprehensive comparison between the VolComm system and the previous method of using Facebook groups for volunteer recruitment. The primary objective of this evaluation was to assess how effectively VolComm enables volunteers to find opportunities that align with their interests and preferences. Respondents rated various aspects of both systems, including ease of use, time efficiency, and overall satisfaction. The findings indicate a strong preference for VolComm, as it offers a more structured and user-friendly experience compared to the disorganized nature of Facebook groups. The system simplifies the process of searching for volunteer activities, making it easier for users to find suitable opportunities. In conclusion, the results suggest that VolComm significantly improves volunteer management by providing an efficient and engaging platform for both volunteers and organizers.

Volunteer Survey Analysis

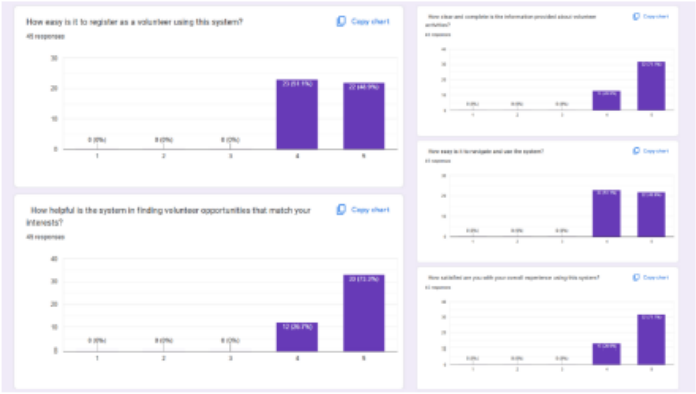


Figure 3: Volunteer Survey Result

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e-ISSN: XXXX-XXXX
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The survey conducted among 45 volunteers revealed overwhelmingly positive feedback about their experience using VolComm. The registration process was rated as easy, with 51.1% of respondents giving it a 4, and 48.9% rating it a 5, showing that all volunteers found the sign-up process smooth and accessible. Similarly, VolComm's ability to match volunteers with suitable opportunities was highly praised, with 73.3% of respondents rating it 5, and 26.7% giving it a 4, demonstrating its effectiveness in connecting users with relevant volunteer activities.

The clarity of information provided about volunteer events was another key factor evaluated. 71.1% of respondents rated this aspect 5, while 28.9% gave it a 4, indicating that the platform provides well-organized and detailed information about activities. Additionally, the user experience and navigation of the system received similarly high ratings, confirming that VolComm is structured for ease of use. The final question, regarding overall satisfaction, showed that 100% of volunteers had a positive experience, with 71.1% rating it 5 and 28.9% rating it 4. These results confirm that VolComm successfully meets volunteers' needs by offering an intuitive and efficient platform for finding and registering for opportunities.

Organizer Survey Analysis

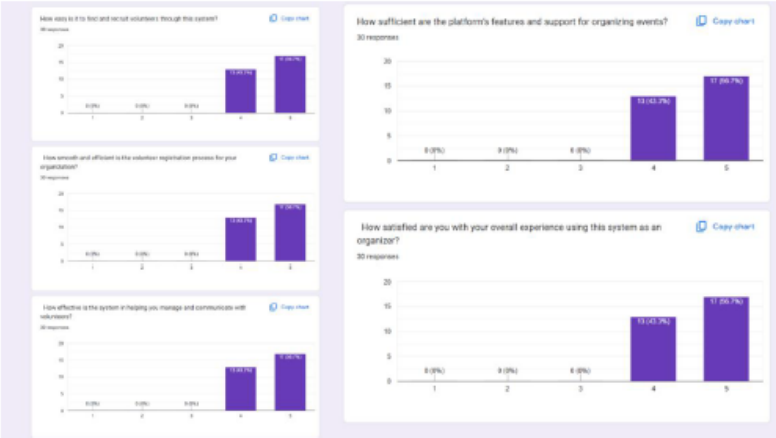


Figure 4: Organizer Survey Result



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A separate survey was conducted among 30 organizers to evaluate their experience using VolComm. The results showed a similarly strong approval of the system. When asked about how easy it was to find and recruit volunteers, 56.7% of organizers rated it 5, while 43.3% gave it a 4, confirming that VolComm effectively streamlines the volunteer recruitment process.

Regarding the efficiency of the volunteer registration process, 100% of organizers rated it either 4 or 5, highlighting that the system ensures a smooth and structured registration flow. Additionally, when asked about the effectiveness of communication and volunteer management features, 56.7% of organizers rated it 5, while 43.3% rated it 4, showing that VolComm provides valuable tools for interacting with volunteers.

The event management support provided by VolComm was also rated highly, with the same 4 and 5 distribution across all responses, indicating that organizers find the system's features sufficient for planning and managing volunteer events. Lastly, the overall satisfaction rating among organizers reflected complete approval, with 100% of respondents giving a 4 or 5, confirming that VolComm successfully addresses their needs and enhances their volunteer coordination processes.

CONCLUSION

VolComm successfully solved the problem of inefficient volunteer management and the challenges faced by both volunteers and organizers in finding and managing volunteer opportunities. Based on the findings, 73.3% of volunteers rated 5 out of 5 in terms of how helpful VolComm is in finding volunteer opportunities that match their interests, while the remaining 26.7% rated it 4 out of 5, proving that the system effectively streamlines the search process. Additionally, 71.1% of volunteers rated 5 out of 5 for the clarity and completeness of information about volunteer activities, with 28.9% rating it 4 out of 5, ensuring that they could make well-informed decisions before participating.

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From the organizer's feedback, 56.7% of organizers rated 5 out of 5 for the ease of finding and recruiting volunteers, while 43.3% rated it 4 out of 5, demonstrating that VolComm significantly improves volunteer outreach. Similarly, 56.7% of organizers rated 5 out of 5 for the system's ability to enhance communication and volunteer management, with 43.3% rating it 4 out of 5, confirming that it effectively reduces administrative workload. The consistently high ratings of 4 out of 5 and 5 out of 5 across all aspects highlight that VolComm has successfully enhanced volunteer participation, simplified event coordination, and improved the overall volunteering experience for both volunteers and organizers.

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PCMJ
Progress in Computer and Mathematics Journal (PCMJ)
volume 1, issues 1 [March, 2024] pp. x
e-ISSN: XXXX-XXXX
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Progress in Computer and Mathematics Journal (PCMJ)
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Best of luck for your FYP

May Allah ease all process and all of you succeed with flying colours.

THANk YOU !



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